

ORACLE: Finance-Informed Expert Routing and Online Abstention for Cross-Sectional Stock Ranking

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Abstract

Cross-sectional stock ranking underpins systematic equity selection, but financial signals are nonstationary. Informative horizons vary across stocks and dates, while trend and reversal rotate across regimes. Time-series forecasters capture individual dynamics but ignore relative movements. Cross-sectional rankers score stocks jointly, yet still tend to fix the lookback horizon and deploy predictions without adapting to the prevailing market state. Both become brittle as regimes shift.

We introduce ORACLE (*Online Routing with Adaptive Calibration and Lookback Experts*), which adapts temporal scale, regime composition, and prediction confidence. ORACLE decomposes each stock’s signal into trend and reversion components, each processed by a separate FactorMoE. Four parallel Bayesian lookback experts spanning 8 to 64 trading days yield a sample-specific predictive mixture. Their allocations align with Barra-style risk exposures without factor supervision, making ORACLE, to our knowledge, the first cross-sectional ranker to learn temporal scale per sample. RegimeMoE then combines the component forecasts: a Feature Gate conditions the mixture on the current market state, while an Online Hedge Gate updates weights using settled losses, tracking regime rotations without retraining or look-ahead. Finally, an Online Abstention Layer updates its threshold from realized errors, tracking regime-driven calibration drift while keeping the abstention rate near its target to preserve portfolio breadth and robustness.

Under matched training budgets and transaction costs, ORACLE achieves the highest mean rank IC and net-of-cost Sharpe ratio among 18 machine-learning, time-series, and cross-sectional baselines across four markets, and is the only method with positive net Sharpe in every market. A controlled label-geometry study further holds model capacity fixed while varying label spacing, showing that spacing rather than architecture determines whether capacity is directed toward rank IC or portfolio Sharpe.

Introduction

Systematic equity strategies allocate capital by comparing opportunities across stocks, which requires more than accurate individual return forecasts. A ranker must identify the useful history, choose between trend and reversion, and avoid trading when confidence is low. These decisions cannot be fixed in advance because markets change.

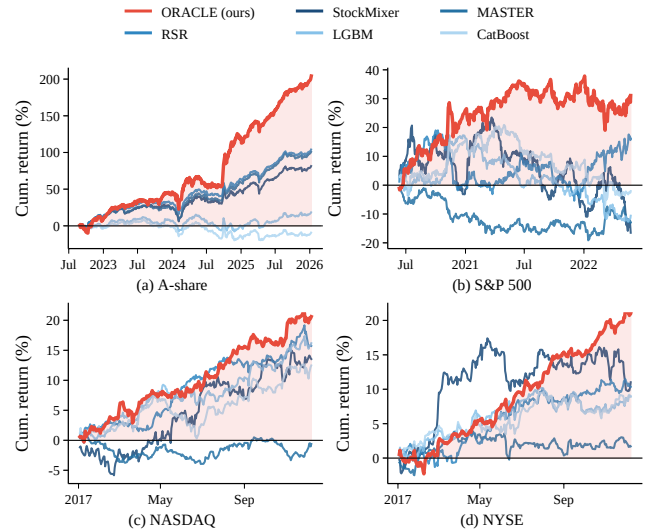


Figure 1: Cumulative NAV over the test period. A-shares: long-only top decile. US markets: gross long–short decile spread. ORACLE in red, baselines in blue.

General time-series forecasters provide strong temporal encoders but continue to treat each asset as a separate prediction problem. Patch, frequency, and decomposition models capture rich structure within a series (Nie et al. 2023; Zhou et al. 2022; Zeng et al. 2023). Pathformer (Chen et al. 2024) also chooses among temporal scales based on the input. These designs are useful for forecasting a series, but an equity portfolio depends on the ordering of many stocks on the same day. A better temporal encoder alone does not resolve that mismatch.

Cross-sectional rankers bring stocks into a shared objective. RSR (Feng et al. 2019) propagates information through a curated relation graph. MASTER (Li et al. 2024) uses a market-status vector for feature selection. StockMixer (Fan and Shen 2024) mixes indicators, time patches, and stocks. These models improve relative representation, but the main routing decisions are still learned once during training. The relation graph is supplied in advance, market conditioning depends on current inputs, and temporal mixing is not revised after a prediction settles. A choice that worked in the training

period may therefore linger after its economic relevance has faded.

Two limitations remain. First, cross-sectional rankers usually retain a lookback design chosen before training, even though the useful horizon can differ across stocks and regimes. Second, market conditioning ends when the prediction is issued. The model does not revise its regime preference after observing which signal actually performed well. This rigidity also affects trade selection. A threshold calibrated on a historical split can drift as the error distribution changes, causing the number of withheld stocks to move far from its intended level (Geifman and El-Yaniv 2019; Sanderink 2026).

ORACLE, short for *Online Routing with Adaptive Calibration and Lookback Experts*, addresses these limitations with feedback at each decision. It chooses temporal scale per stock-day, updates trend-reversion balance from observed performance, and adapts abstention while stabilizing portfolio breadth. Outcomes enter online updates only after their returns settle.

ORACLE separates each stock-day into trend and reversion components, then sends each component to its own **FactorMoE**. The gate chooses among four Bayesian convolutional branches spanning 8 to 64 trading days. As a result, different stocks and dates can rely on different amounts of history. **RegimeMoE** combines the two component forecasts. Its Feature Gate uses the current window, whereas its Online Hedge Gate uses the accumulated losses of the two experts. One reacts to the market now and the other records which dynamic has worked in practice. The mixture can change as regimes rotate without retraining or using future information.

The same feedback principle governs trade selection. Predictive uncertainty from the Bayesian branches enters an **Online Abstention Layer**, which adjusts its threshold as realized errors arrive. This keeps the number of withheld stocks close to the intended level when calibration drifts, preserving portfolio breadth. The routing decisions remain interpretable as well. Even without factor supervision, the learned horizon weights align with Barra-style risk exposures.

We evaluate ORACLE against 18 baselines on four markets using matched training budgets and transaction costs. It achieves the highest mean rank IC and net-of-cost Sharpe and is the only method with positive net Sharpe in every market. We also vary label spacing while holding the model, data, training, and supervised ordering fixed. Rank IC and portfolio Sharpe favor opposite parts of this sweep. The result shows label design can determine where model capacity is spent even when the architecture does not change.

Contributions.

- **We propose FactorMoE**, which learns a mixture of Bayesian lookback experts for each sample, and **RegimeMoE**, which combines current market features with Hedge updates from settled losses. Together they adapt temporal scale and the balance between trend and reversion without retraining. The learned weights also align with Barra-style risk exposures without factor supervision.
- **We introduce an Online Abstention Layer** for cross-

sectional ranking that learns from daily realized errors to control the abstention rate through regime shifts, stabilizing portfolio breadth while improving the quality of retained trades.

- **We isolate** the effect of label geometry. With architecture, data, and training fixed, label spacing determines whether model capacity is directed toward rank IC or portfolio Sharpe.
- **We evaluate** 18 machine-learning, time-series, and cross-sectional baselines on four markets (CSI 300, S&P 500, NASDAQ, NYSE) under a unified protocol. ORACLE achieves the best mean rank IC and net-of-cost Sharpe and is the only model with positive net Sharpe on every market.

Related Work

Temporal and cross-sectional stock modeling. General forecasters model within-series structure through decomposition (Zeng et al. 2023), patching (Nie et al. 2023), inverted attention (Liu et al. 2024), periodic representations (Wu et al. 2023; Zhou et al. 2022), feature mixing (Chen et al. 2023; Zhang et al. 2022), and multi-scale pathways (Chen et al. 2024). These methods are designed to predict each series, whereas stock selection requires returns to be ordered within the same-day universe. Cross-sectional models address this distinction through relation graphs (Feng et al. 2019), higher-order co-movement (Sawhney et al. 2021), market-guided feature selection (Li et al. 2024), and mixing across stocks (Fan and Shen 2024). Their historical context, however, is still chosen before training and then shared across stock-days. A horizon that works on average in the training period need not remain appropriate for stocks with different exposures or during a later regime. Existing work has not examined this variation within a cross-sectional ranking objective.

Learning and deployment under market drift. Mixture-of-experts models route observations according to a learned gate (Jacobs et al. 1991; Shazeer et al. 2017), and MIGA brings input-dependent expert aggregation to stock prediction (Yu et al. 2024). Once training ends, the routing rule is fixed and subsequent decisions depend on current inputs alone. Realized prediction losses do not revise which expert should be trusted after the market changes. Online expert aggregation weights predictors by their past losses (Freund and Schapire 1997; Cesa-Bianchi and Lugosi 2006), but its regret analysis concerns aggregate sequential prediction. It does not distinguish persistent market regimes from heterogeneous stock states. In cross-sectional ranking, these two sources of variation are easily conflated: current features may reflect short-lived differences even as settled outcomes reveal a broader regime rotation.

Monte Carlo dropout and mixture density networks estimate epistemic and aleatoric uncertainty (Gal and Ghahramani 2016; Bishop 1994). Selective prediction then trades coverage for accuracy by withholding uncertain outputs (Geifman and El-Yaniv 2017, 2019). Conformal methods provide coverage guarantees under exchangeability (Vovk, Gammerman, and Shafer 2005; Angelopoulos and Bates 2023), while adaptive conformal inference extends cover-

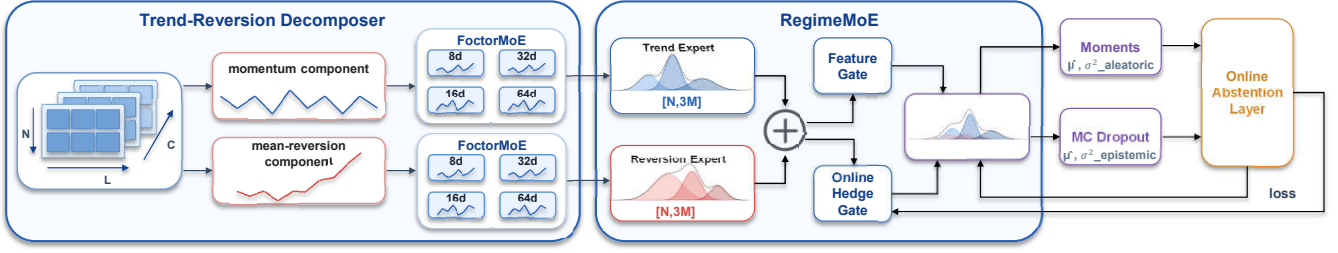


Figure 2: The ORACLE pipeline. FactorMoE mixes four Bayesian branches spanning 8 to 64 days per component. RegimeMoE recombines components via Feature Gate and Online Hedge Gate. The Expert Mixture Head outputs a predictive distribution fed to the Online Abstention Layer. Dashed arrows carry settled-loss feedback.

age control to distribution shift (Gibbs and Candès 2021). The target remains predictive coverage rather than the composition of an investment portfolio. In finance, Sanderink (2026) uses uncertainty for a strategy-level trust gate and a position-level tail-risk cap, but does not study stock-level abstention or portfolio breadth. The unresolved deployment problem is therefore not coverage alone. When calibration drifts, a fixed cutoff can withhold sharply different numbers of stocks across regimes, changing diversification and risk even if the underlying ranking is unchanged.

Supervision and financial evaluation. Training loss affects the ranking learned by a stock model (Kwiatkowski and Chudziak 2025), and the best supervision horizon need not match the forecast horizon (Song, Liu, and Chen 2026). Evaluation choices matter as well. Recent benchmark evidence shows that model rankings change with market state, transaction costs, portfolio constraints, and investor preferences (Zhang, Cheng, and Leung 2026). Existing work treats loss choice, horizon choice, and evaluation protocol as separate design decisions. It does not isolate label spacing while keeping the supervised ordering, model, data, and training procedure fixed. The effect of label geometry therefore remains confounded with model and evaluation choices.

Method

Problem Formulation and Pipeline

On day t , the market provides $X_t \in \mathbb{R}^{N_t \times L \times F}$ for N_t stocks, with L days of F price-volume features per stock. Each window is instance-normalized. The target is the next-day return standardized within the live cross-section,

$$y_{i,t} = \frac{r_{i,t+1} - \bar{r}_t}{\sigma_t}, \quad (1)$$

following Li et al. (2024). The model produces a score for every stock, and the scores are trained jointly over the same-day universe.

Figure 2 summarizes the pipeline. ORACLE first separates each window into trend and reversion components. Two independent **FactorMoE** towers then decide how much history each component needs. **RegimeMoE** combines their predictive distributions using current features and settled expert losses. The resulting distribution supports both the ranking loss and the **Online Abstention Layer**. This order matters:

temporal evidence is selected within each economic component before the model decides which regime to emphasize.

Finance-Informed Dual-Tower FactorMoE

Traditional factor construction evaluates related signals over several formation windows, then chooses their lengths and combinations from historical experience. A single fixed look-back hard-codes that judgment for every stock-day. It is especially restrictive when trend persistence and short-term reversal draw information from different parts of the history (Lehmann 1990; Jegadeesh and Titman 1993).

We begin with a moving-average decomposition of each stock window:

$$X_{i,t}^{\text{tr}} = \text{MA}_k(X_{i,t}), \quad X_{i,t}^{\text{rev}} = X_{i,t} - X_{i,t}^{\text{tr}}. \quad (2)$$

The two components have different statistical structure, so they do not share an encoder. Each component $c \in \{\text{tr}, \text{rev}\}$ receives its own FactorMoE with four Bayesian convolutional branches. The scale set $\mathcal{S} = \{8, 16, 32, 64\}$ is an approximately geometric collection of short-to-medium financial horizons rather than an exhaustive search over window lengths. For stock i on day t ,

$$\begin{aligned} e_{s,i,t}^{(c)} &= g_s^{(c)}(X_{i,t-s+1:t}^{(c)}), \\ w_{i,t}^{(c)} &= \text{softmax}\left(\text{MLP}^{(c)}\left([e_{8,i,t}^{(c)}; e_{16,i,t}^{(c)}; e_{32,i,t}^{(c)}; e_{64,i,t}^{(c)}]\right)\right), \\ e_{i,t}^{(c)} &= \sum_{s \in \mathcal{S}} w_{i,t,s}^{(c)} e_{s,i,t}^{(c)}. \end{aligned} \quad (3)$$

The branches produce temporal representations, not separate return forecasts. Their gate weights expose the horizon allocation chosen for each component and stock-day. Table 3 tests whether adaptive allocation matters, while Figure 3 relates the learned allocations to Barra-style exposures that were never supplied to the gate.

Finance-Informed RegimeMoE

Trend and reversion are not merely two labels on the same representation. Their useful histories can differ, which is why the two FactorMoEs in Section receive different inputs and learn separate scale mixtures. A **Trend Expert** and a **Reversion Expert** map the resulting embeddings to Gaussian mixtures $p_c(y | X) = \sum_{m=1}^M \omega_m^{(c)} \mathcal{N}(y; \mu_m^{(c)}, (\sigma_m^{(c)})^2)$.

RegimeMoE combines these component distributions with two sources of evidence. The **Feature Gate** maps break-out amplitude, trend strength, volatility state, and momentum descriptors from the current window to logits $f_{i,t} \in \mathbb{R}^2$. This gate reacts immediately, but its parameters are learned from the training period. The **Online Hedge Gate** instead records which expert has performed well after outcomes settle. For expert $k \in \{1, 2\}$,

$$\begin{aligned}\tilde{v}_{k,t+1} &= \frac{v_{k,t} \exp(-\eta \ell_{k,t})}{\sum_j v_{j,t} \exp(-\eta \ell_{j,t})}, \\ v_{k,t+1} &= (1 - \beta_{\text{fs}}) \tilde{v}_{k,t+1} + \frac{\beta_{\text{fs}}}{2}, \\ \pi_{i,t} &= \text{softmax}(f_{i,t} + \log v_t).\end{aligned}\quad (4)$$

Here $\ell_{k,t}$ is the settled NLL of expert k . The first line is the Hedge update (Freund and Schapire 1997; Cesa-Bianchi and Lugosi 2006). The Fixed-Share step (Herbster and Warmuth 1998) keeps a nonzero path back to either regime after a long run dominated by the other. Adding $\log v_t$ to the Feature Gate logits is equivalent to multiplying its probabilities by the online weights. The two gates therefore operate on different evidence: current features explain stock-level variation, while settled losses reveal persistent changes in expert performance. Only losses whose return horizon has elapsed enter Equation (4). Table 4 evaluates the contribution of the components through a staged build-up.

Rank-Aligned Distributional Objective

RegimeMoE produces a mixture with $2M$ Gaussian components. We train this distribution with tail-weighted negative log-likelihood and a correlation penalty. Let $z_i = |y_i| / (\text{Std}_{j \in B} |y_j| + \epsilon)$ for batch B . The objective is

$$\mathcal{L} = \underbrace{\frac{1}{B} \sum_{i=1}^B (1 + \lambda_{\text{tail}} z_i) \text{NLL}_i}_{\text{distributional term}} + \lambda_{\text{ic}} (1 - \rho(\hat{\mu}, y)), \quad (5)$$

where ρ is the in-batch Pearson correlation between predicted means $\hat{\mu}$ and labels y , with $\lambda_{\text{ic}}=8$ and $\lambda_{\text{tail}}=0.5$. The first term preserves a predictive distribution and gives large moves adequate weight. The second aligns its conditional mean with the cross-sectional ranking objective.

Dropout remains active in every FactorMoE branch at inference. Ten stochastic passes give

$$\hat{\sigma}_a^2 = \mathbb{E}_k[\text{Var}(Y \mid \theta_k, X)], \quad \hat{\sigma}_e^2 = \text{Var}_k[\mathbb{E}(Y \mid \theta_k, X)], \quad (6)$$

where $\hat{\sigma}_a^2$ captures aleatoric uncertainty and $\hat{\sigma}_e^2$ captures epistemic uncertainty (Gal and Ghahramani 2016; Kendall and Gal 2017). Operationally, $\hat{\sigma}_a^2$ averages the conditional variance emitted by the Gaussian mixtures across stochastic passes, while $\hat{\sigma}_e^2$ measures how much their predictive means disagree under MC dropout. The former is large when a stock’s return is intrinsically noisy even under a stable model. The latter grows when plausible model parameters produce different forecasts, as can happen in an unfamiliar market state. Their sum $\hat{\sigma}^2 = \hat{\sigma}_a^2 + \hat{\sigma}_e^2$ provides the risk scale used by the Online Abstention Layer.

Online Abstention Layer under Distribution Shift

A forecast with large uncertainty relative to its expected return should not automatically become a position. A fixed uncertainty cutoff is also insufficient. When the scale of predictive noise shifts, the same cutoff can withhold too few stocks in one regime and too many in another. The **Online Abstention Layer** therefore separates error calibration from portfolio-breadth control.

The calibration step uses only settled outcomes. After the return for stock i on day t is observed, its standardized non-conformity score is $s_{i,t} = |r_{i,t} - \hat{\mu}_{i,t}| / \hat{\sigma}_{i,t}$. Adaptive conformal inference (Gibbs and Candès 2021) updates the mis-coverage level and the trailing error quantile:

$$\begin{aligned}\text{err}_t &= \frac{1}{N_t} \sum_{i=1}^{N_t} \mathbf{1}\{s_{i,t} > q_t\}, \\ \alpha_{t+1} &= \alpha_t + \eta_c (\alpha^* - \text{err}_t), \\ q_{t+1} &= \text{Quantile}_{1-\alpha_{t+1}}(\mathcal{B}_{t-W+1:t}),\end{aligned}\quad (7)$$

where $\mathcal{B}_{t-W+1:t}$ contains settled scores from the trailing window. Thus q_t tracks changes in the realized error distribution rather than remaining tied to one calibration split.

Calibration alone does not determine how many stocks are withheld. Before the day- t return is known, the layer forms the confidence ratio

$$\begin{aligned}c_{i,t} &= \frac{|\hat{\mu}_{i,t}|}{q_t \hat{\sigma}_{i,t}}, \\ a_t &= \frac{1}{N_t} \sum_{i=1}^{N_t} \mathbf{1}\{c_{i,t} < \tau_t\}, \\ \tau_{t+1} &= \tau_t \exp[\eta_a (a^* - a_t)].\end{aligned}\quad (8)$$

Stocks with $c_{i,t} < \tau_t$ are withheld, and the portfolio is formed on the remaining cross-section. If the observed abstention rate a_t falls below its target a^* , the threshold rises for the next decision. If too many stocks are withheld, it falls. Equation (7) learns from settled errors, while Equation (8) responds to the inference-time confidence distribution. No day- t return affects the day- t portfolio. Figure 4 reports calibration under drift, and the technical appendix gives the corresponding online guarantees.

Experiments

Setup

Markets and protocol. We evaluate four cross-sections: **A-share** ($5,976$ stocks $\times 4,138$ days), **S&P 500** ($474 \times 2,526$), **NASDAQ** ($1,026 \times 1,245$), and **NYSE** ($1,737 \times 1,245$). All methods use the same five-feature market panels, chronological 70/10/20 splits, cross-sectionally standardized labels, and a 300s training budget on one NVIDIA H800 GPU. Cross-sectional rankers additionally receive the auxiliary inputs required by their reference implementations. We report five-seed results for ORACLE, cross-sectional rankers, and gradient-boosting models, and three-seed results for time-series forecasters. No model is selected using test performance.

Model	A-share		S&P 500		NASDAQ		NYSE	
	RIC	ICIR	RIC	ICIR	RIC	ICIR	RIC	ICIR
<i>General time-series forecasters (3 seeds each)</i>								
DLinear	0.18±0.16	0.016±0.014	0.34±0.29	0.020±0.017	0.28±0.08	0.029±0.008	-0.04±0.11	-0.005±0.012
PatchTST	0.65±0.25	0.079±0.032	-0.67±0.22	-0.055±0.019	0.35±0.26	0.049±0.034	0.70±0.14	0.106±0.022
iTransformer	1.14±0.69	0.138±0.085	-0.11±0.28	-0.005±0.018	0.00±0.37	0.009±0.048	0.77±0.32	0.091±0.035
TimesNet	1.76±0.94	0.146±0.064	0.05±0.43	0.003±0.025	0.25±0.34	0.022±0.030	-0.08±0.41	-0.007±0.032
TSMixer	0.97±0.20	0.122±0.028	0.30±0.31	0.028±0.029	0.56±0.06	0.088±0.008	0.77±0.09	0.129±0.015
Pyraformer	1.87±0.46	0.208±0.056	-0.19±0.16	-0.013±0.012	1.09±0.35	0.121±0.040	-0.15±0.35	-0.019±0.041
Crossformer	1.02±1.39	0.117±0.137	-0.24±0.09	-0.015±0.006	0.34±0.20	0.032±0.018	0.21±0.43	0.019±0.040
FEDformer	0.13±1.21	0.022±0.126	0.13±0.61	0.009±0.042	0.46±0.50	0.049±0.051	0.36±0.62	0.032±0.060
FiLM	-0.27±1.71	-0.071±0.220	-0.05±0.03	-0.010±0.007	-0.05±0.26	-0.006±0.025	0.07±0.60	0.017±0.053
FreTS	0.70±0.16	0.101±0.022	0.15±0.23	0.019±0.029	0.28±0.14	0.044±0.022	0.44±0.11	0.084±0.024
LightTS	0.16±0.56	0.017±0.061	-0.14±0.53	-0.011±0.039	0.76±0.10	0.086±0.018	0.48±0.50	0.059±0.061
SCINet	0.23±1.23	0.028±0.127	-0.54±0.43	-0.033±0.025	0.36±0.27	0.040±0.031	-0.09±0.74	-0.001±0.073
TiDE	0.16±0.41	0.008±0.046	0.20±0.40	0.012±0.024	-0.28±0.30	-0.028±0.030	0.20±0.40	0.021±0.042
<i>Gradient-boosting baselines (5 seeds each)</i>								
LGBM	-2.84±0.10	-0.226±0.008	-0.42±0.01	-0.036±0.001	1.00±0.08	0.095±0.009	1.87±0.08	0.160±0.006
CatBoost	-3.04±0.05	-0.227±0.004	-0.24±0.01	-0.015±0.001	1.15±0.12	0.105±0.011	1.65±0.08	0.143±0.008
<i>Cross-sectional rankers (5 seeds each)</i>								
StockMixer	3.76±0.55	0.341±0.037	-0.78±0.55	-0.038±0.027	1.06±0.61	0.081±0.047	0.31±0.43	0.021±0.028
MASTER	4.91±0.35	0.475 ±0.038	0.19±0.24	0.018±0.025	-0.33±0.43	-0.050±0.066	-0.03±0.46	0.004±0.068
RSR	4.63±0.59	0.471±0.048	0.35±0.33	0.022±0.021	1.36±0.29	0.170±0.049	1.46±0.35	0.139±0.031
<i>Ours (5 seeds)</i>								
ORACLE	4.95 ±0.73	0.464±0.056	0.53 ±0.52	0.030 ±0.030	2.13 ±0.60	0.211 ±0.062	2.40 ±0.84	0.180 ±0.064

Table 1: Daily rank IC ($\times 10^2$) and rank ICIR as mean $\pm$ std under identical budget, splits, loss, and labels. Bold marks the column best.

Evaluation and baselines. Rank IC and its information ratio measure full-universe ordering, while cost-adjusted decile Sharpe measures the performance of the traded extremes. We use long-only top-decile portfolios for A-shares and long-short top-minus-bottom deciles for US markets. ORACLE is compared with 18 baselines: three cross-sectional rankers, two gradient-boosting models, and thirteen time-series forecasters. Dataset construction, input windows, transaction costs, the complete baseline list, and implementation details are provided in the technical appendix.

Main Results and Analysis

Table 1 reports ranking quality and Table 2 the corresponding portfolio backtests. These evaluations answer different questions. Rank IC uses the full cross-section, whereas decile Sharpe asks whether the stocks at the traded extremes remain useful after turnover and costs.

ORACLE’s clearest advantage is consistency across markets. It records the highest mean rank IC in all four universes, at 4.95 on A-shares, 0.53 on S&P 500, 2.13 on NASDAQ, and 2.40 on NYSE. The strongest baseline changes with the market. MASTER nearly matches ORACLE on A-shares at 4.91, RSR is second on S&P 500 and NASDAQ at 0.35 and 1.36, and LGBM is second on NYSE at 1.87. ORACLE’s margin is consequently small on A-shares but widens to 0.77 rank IC points on NASDAQ and 0.53 on NYSE. Rank ICIR gives a similar but more qualified picture. ORACLE leads on all three US markets, while MASTER leads A-shares

by 0.011. The result is therefore a four-market advantage in mean ordering quality, not a claim that ORACLE wins every reported statistic.

The same ordering quality remains investable after costs. ORACLE leads every gross and net Sharpe column in Table 2. Its Sharpe changes from 2.70 to 2.11 on A-shares, 0.45 to 0.08 on S&P 500, 3.71 to 2.64 on NASDAQ, and 2.88 to 2.00 on NYSE after costs. The S&P 500 result is the thinnest edge, but every reported baseline becomes negative after costs in that market. The separation is much larger on NASDAQ and NYSE, where the strongest baseline net Sharpes are 0.90 and 0.70. ORACLE is thus the only method with positive net Sharpe in every market. Since it already leads the gross columns, this advantage is not created by the cost model.

Figure 1 shows that the portfolio result is not confined to the final observation. ORACLE separates clearly from the comparison methods on A-shares, NASDAQ, and NYSE, although the path remains volatile and the S&P 500 advantage is narrower. The baseline families are similarly market-sensitive. Tree models post negative rank IC on A-shares and S&P 500 but competitive results on NASDAQ and NYSE, independent time-series forecasters succeed in one universe, and cross-sectional rankers change order across markets. This pattern favors retaining both within-stock temporal structure and same-day cross-sectional context, then adapting how the two are used.

Model	A-sh (long)		SPX (L-S)		NDQ (L-S)		NYSE (L-S)	
	gross	net	gross	net	gross	net	gross	net
LGBM	0.49	-0.28	-0.43	-1.35	1.51	0.48	1.64	0.70
CatBoost	-0.20	-0.88	-0.07	-0.77	1.33	0.44	1.41	0.47
StockMixer	1.99	1.21	-0.72	-1.25	1.07	0.02	0.21	-0.88
MASTER	2.42	1.66	0.08	-0.85	-0.58	-2.35	0.27	-1.53
RSR	2.62	1.82	0.06	-0.63	2.55	0.90	1.72	0.36
ORACLE	2.70	2.11	0.45	0.08	3.71	2.64	2.88	2.00

Table 2: Decile backtests, annualized Sharpe, 5-seed means. A-shares: long-only top decile. US markets: long-short spread. Net Sharpe deducts real costs per unit of turnover (A-share 10.2bps, US 2.78bps). Bold marks the column best.

Ablation and Analysis

Sample-specific lookback selection consistently improves FactorMoE. Table 3 compares the FactorMoE routing in Section with two controls. Uniform Mixing retains all four temporal branches with fixed weights, testing whether multiple horizons alone suffice. Fixed 1-Day removes the temporal-scale bank, testing the cost of commitment to one short history. FactorMoE leads both controls in every market. Relative to Uniform Mixing, the gains are 1.48 rank IC points on A-shares, 0.40 on S&P 500, 0.07 on NASDAQ, and 0.19 on NYSE. Relative to Fixed 1-Day, they are 0.76, 0.84, 0.31, and 0.11.

Fixed 1-Day’s largest deficit on S&P 500 confirms the cost of a short window. Because Uniform Mixing retains every branch, its A-share gap reflects routing rather than multi-scale capacity. The gains are smaller on NASDAQ and NYSE but positive throughout. Thus, useful horizons vary by stock and date, while adaptation gains remain market-dependent.

Horizon mode	A-sh	SPX	NDQ	NYSE
FactorMoE (Ours)	4.95 (0.7)	0.53 (0.5)	2.13 (0.6)	2.94 (0.4)
Uniform Mixing	3.47 (1.1)	0.13 (0.4)	2.06 (0.5)	2.75 (0.4)
Fixed 1-Day	4.19 (0.6)	-0.31 (0.5)	1.82 (1.2)	2.83 (0.4)

Table 3: Adaptive-lookback ablation, rank IC $\times 10^2$ over 5 seeds with std in parentheses. Bold marks the column best.

FactorMoE learns economically interpretable scale allocations. Figure 3 relates learned scale weights to risk exposures unseen during training. On A-shares, residual volatility correlates negatively with the 32-day weight at -0.128 and positively with the 16-day weight at $+0.094$. The gate therefore shortens the effective memory as idiosyncratic volatility rises rather than using the same formation period for every stock. On NYSE, momentum exposure shifts weight toward the 8-to-32-day branches and away from the 64-day branch. This suggests the gate distinguishes actionable short-to-medium continuation from slower summaries.

Barra-style variables are post hoc (Menchero, Orr, and Wang 2011) and neither supervise the branches nor enter the Scale Gate. The correlations therefore diagnose selected

temporal evidence, not reconstruction of a Barra model. Together, Table 3 and Figure 3 show that adaptive allocation improves prediction while remaining economically interpretable outside the training objective.

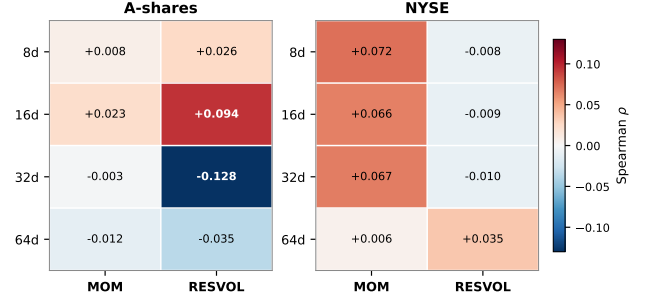


Figure 3: Spearman ρ between per-scale gate weight w_s and Barra-style exposures to momentum (MOM) and residual volatility (RESVOL) over the test period. Red marks routing preference and blue marks avoidance.

The full system trades market-specific peaks for cross-market robustness. Table 4 traces the architecture from a single-scale single-expert Base through to the full ORACLE, measuring net Sharpe over the test set. The build-up is not monotone across markets. FactorMoE is the most specialized stage: it raises NASDAQ from 1.94 to 2.78 and NYSE from 1.68 to 3.47, but lowers A-share and S&P 500 Sharpe. Adding the trend-reversion decomposition and RegimeMoE shifts this profile toward the latter two markets.

Relative to +FactorMoE, full ORACLE gives up 0.14 Sharpe on NASDAQ and 1.47 on NYSE, where temporal-scale routing alone is especially strong. In return, it gains 0.90 on A-shares and 0.27 on S&P 500. It is the only stage with positive net Sharpe in every market, and raises the worst-market result from -0.19 for +FactorMoE to 0.08. Full ORACLE also improves all four columns relative to +RegimeMoE, so the lower NASDAQ and NYSE peaks cannot be assigned to Online ACI alone. The full design favors a balanced cross-market profile over maximizing the best-performing market in isolation.

	A-sh	SPX	NDQ	NYSE
Base	1.63 (0.26)	-0.01 (0.53)	1.94 (0.60)	1.68 (0.14)
+FactorMoE	1.21 (0.36)	-0.19 (0.44)	2.78 (0.45)	3.47 (0.48)
+RegimeMoE	1.74 (0.27)	-0.05 (0.51)	2.42 (0.57)	1.82 (0.61)
ORACLE (full)	2.11 (0.11)	0.08 (0.47)	2.64 (0.63)	2.00 (0.88)

Table 4: Routing build-up from market-specific specialization to cross-market robustness, measured by net Sharpe ratio (mean, std in parentheses). Bold marks the column best.

Online calibration preserves target coverage under distribution shift. Figure 4 holds ORACLE’s predictions fixed and changes only the abstention rule from Section . Before this online step, ORACLE combines two sources of uncertainty. The aleatoric term averages the variance emitted by the mixture-density heads and rises with irreducible return noise.

The epistemic term measures dispersion across MC-dropout predictive means and rises when plausible model parameters disagree. Their sum enters $\hat{\sigma}$, lowering confidence for noisy or unfamiliar stock-days.

Figure 4 then compares calibration rules. The fixed rule reaches 1.00 coverage on all three US markets. Its intervals are too conservative to maintain useful calibration at the target. Static conformal calibration shows the opposite failure under drift. It begins near the target, but S&P 500 coverage falls to 0.85 on average and 0.81 in the worst rolling window once the calibration and test regimes diverge. Online ACI keeps coverage within 0.02 of the 0.90 target across all four markets by updating from settled errors rather than reusing one historical correction.

The three mechanisms have separate roles. Dual uncertainty sets the stock-level risk scale, ACI updates that scale from realized errors, and Equation (8) adjusts the threshold when the withheld fraction departs from its target. This last feedback keeps portfolio breadth under control as calibration changes. Figure 4 validates the calibration step, but does not attribute its gain to either uncertainty component without a separate ablation.

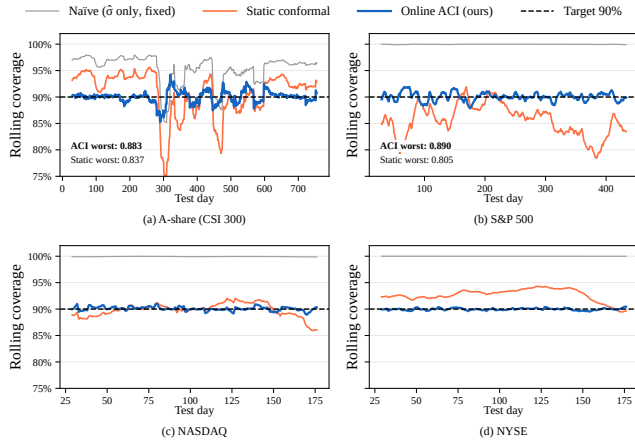


Figure 4: Rolling 30-day empirical coverage on identical ORACLE predictions. The dashed line marks the 90% target. Curves show Fixed Threshold (grey), Static Conformal (orange), and Online ACI (blue, ours), with one panel per market.

The Label Protocol Selects the Metric Axis

Label geometry decides which metric a ranker optimizes, even when the supervised ordering never changes. Every published ranker commits, usually silently, to a label convention: raw returns, z-scored returns, or ranks. We make the convention a continuous dial: $y_\gamma = \text{Norm}(\text{sign}(r) |r|^\gamma)$, where $\gamma=1$ recovers the standard cross-sectional z-score and $\gamma \rightarrow 0$ degenerates to a binary sign label. The transform is strictly monotone for every $\gamma > 0$, so all sweep points supervise the *same* cross-sectional ordering and only the label spacing and tail leverage change. The sweep therefore isolates label geometry from ordinal content.

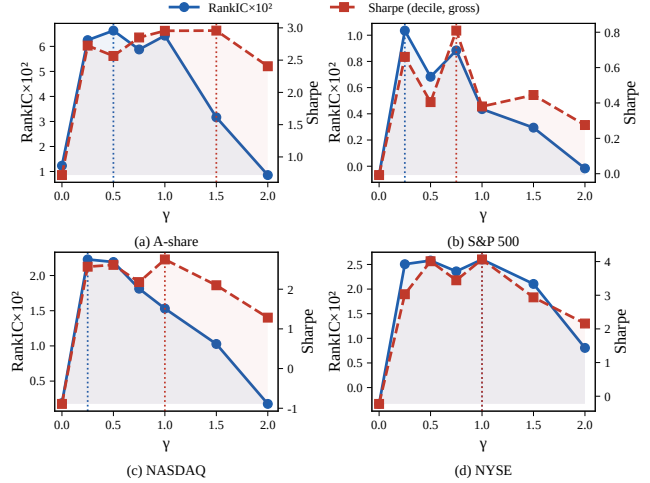


Figure 5: Label transform sweep $y_\gamma = \text{Norm}(\text{sign}(r) |r|^\gamma)$. 2×2 grid: A-share (top-left), S&P 500 (top-right), NASDAQ (bottom-left), NYSE (bottom-right). Blue circles: rank IC ($\times 10^2$). Red squares: gross decile Sharpe. Dotted verticals mark each metric’s peak γ .

With architecture, data, and training held fixed, Figure 5 shows that magnitude-compressed labels with $\gamma < 1$ maximize rank IC while costing portfolio Sharpe, and magnitude-faithful labels with $\gamma \geq 1$ do the reverse. On A-shares, moving from the rank-IC peak at $\gamma=0.5$ to the Sharpe peak at $\gamma=1.5$ halves rank IC from 6.6 to 3.2 while Sharpe rises from 2.6 to 3.0. The same two peaks appear on all four markets.

The implication is that two papers that differ in label convention optimize different objectives, and their rank IC or Sharpe reflect those choices as much as their architectures. Reporting the label convention alongside the model is therefore necessary for meaningful cross-paper comparison.

Conclusion

ORACLE treats cross-sectional stock ranking as an adaptation problem. FactorMoE selects temporal evidence, and RegimeMoE routes between Trend and Reversion Experts before each prediction. After returns arrive, settled expert losses update routing, while the Online Abstention Layer updates its threshold. On four markets (CSI 300, S&P 500, NASDAQ, NYSE) against 18 baselines under an equal-budget protocol, ORACLE achieves the best mean rank IC and net-of-cost Sharpe and is the only model with positive net Sharpe on every market. Ablation shows that the learned horizon weights correlate with Barra-style risk exposures, that no single fixed lookback dominates across all markets, and that online feedback, not exchangeability, is what keeps the abstention budget controlled through regime breaks. A controlled label-geometry sweep further reveals that label spacing, not architecture, determines whether capacity is allocated toward rank IC or Sharpe, underscoring that preprocessing choices are part of the learning problem.

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